

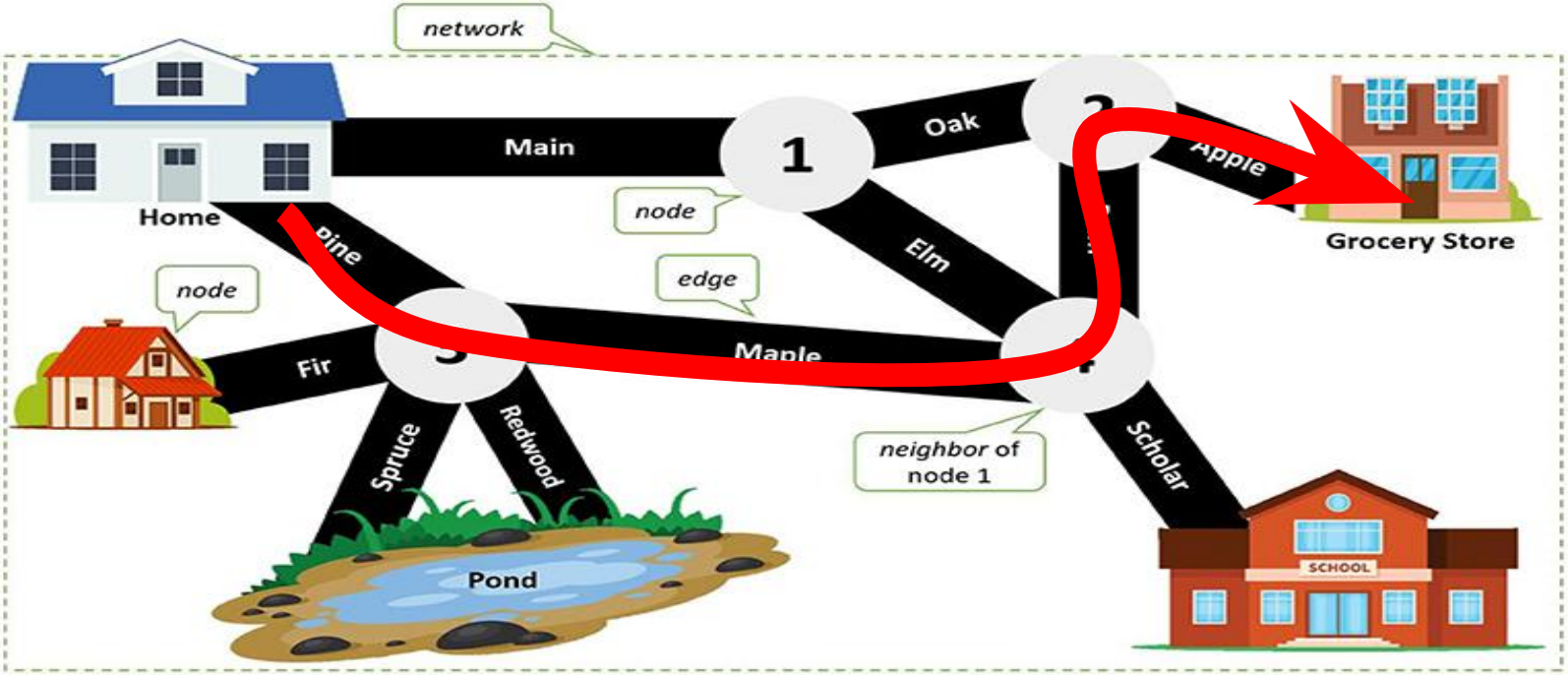
Data Structure(CSL202)

Dr. Sk Subidh Ali (subidh@iitbhilai.ac.in)



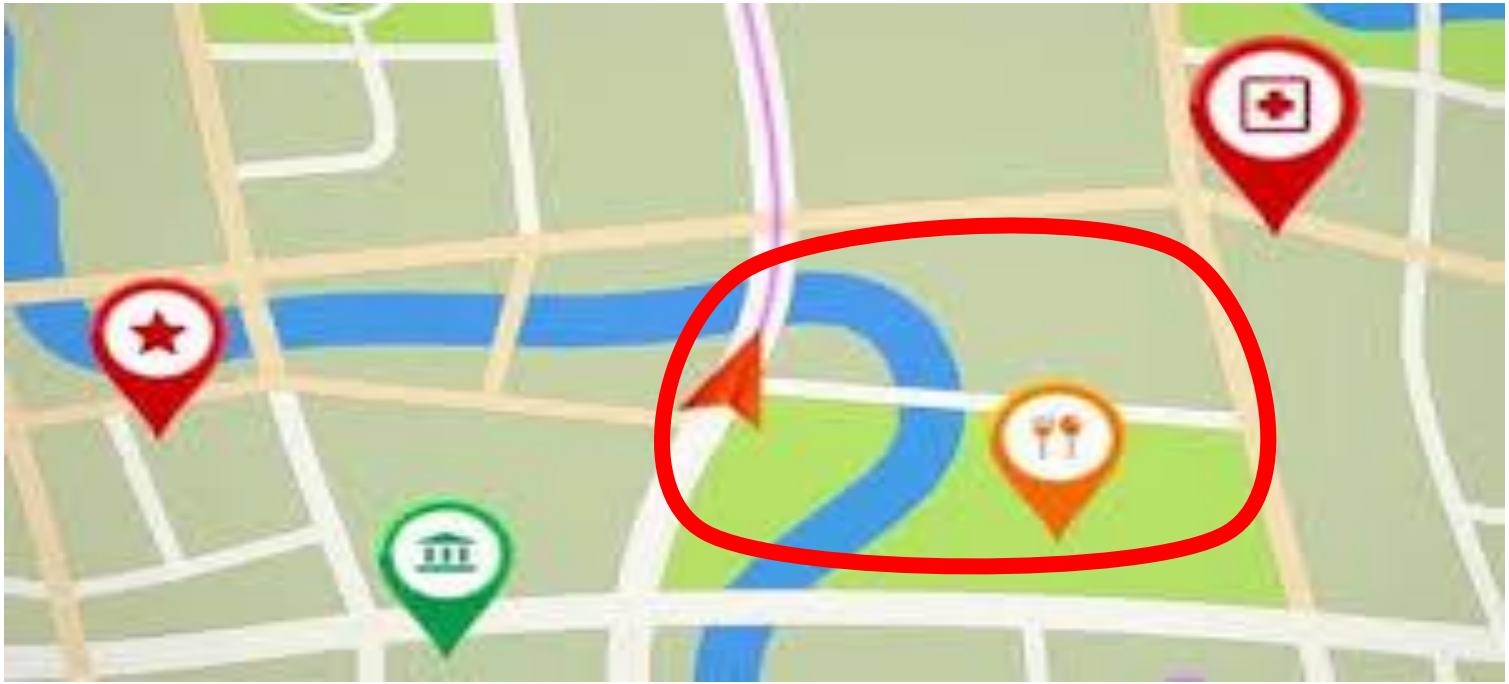
Searching in Graph: DFS and BFS

Why We Need To Search In Graph



Find path between Source and Destination, if it exists

Why We Need To Search In Graph



Detect Cycle

Why We Need To Search In Graph



Find Shortest Path: Navigation, Packet routing in network

Why We Need To Search In Graph



- Friend suggestion in Social networking (BFS/DFS)
- Suggest product or movies in recommendation system



Depth First Search DFS

Depth First Search

- An algorithm, which is used to visit all of the nodes of a given graph.
- First we traverse the starting node A.
- We process an adjacent of A, then adjacent of an adjacent of A, and so on.
- Backtrack when a dead-end is reached.
- Uses a stack to track nodes.

Depth First Search

Algorithm:

1. Start from a selected node (source).
2. Visit and mark it as visited.
3. Move to an adjacent unvisited node.
4. Repeat until you reach a dead-end.
5. Backtrack and explore other unvisited paths.

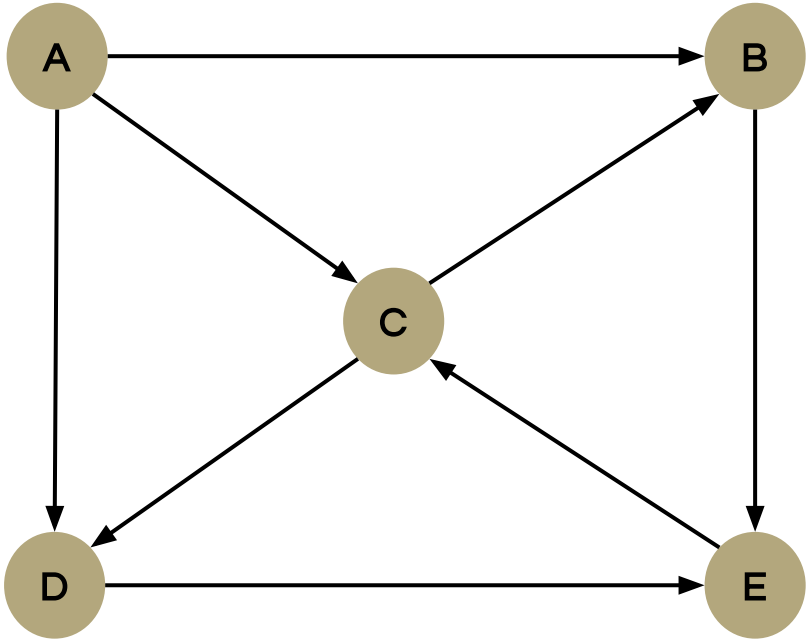
Depth First Search

- **DFS Time Complexity:**
  - Using Adjacency List: O(V + E)
  - (V = vertices, E = edges)
  - Using Adjacency Matrix: $O(V^2)$
- **DFS Space Complexity: O(V)**
  - (for visited array + stack)

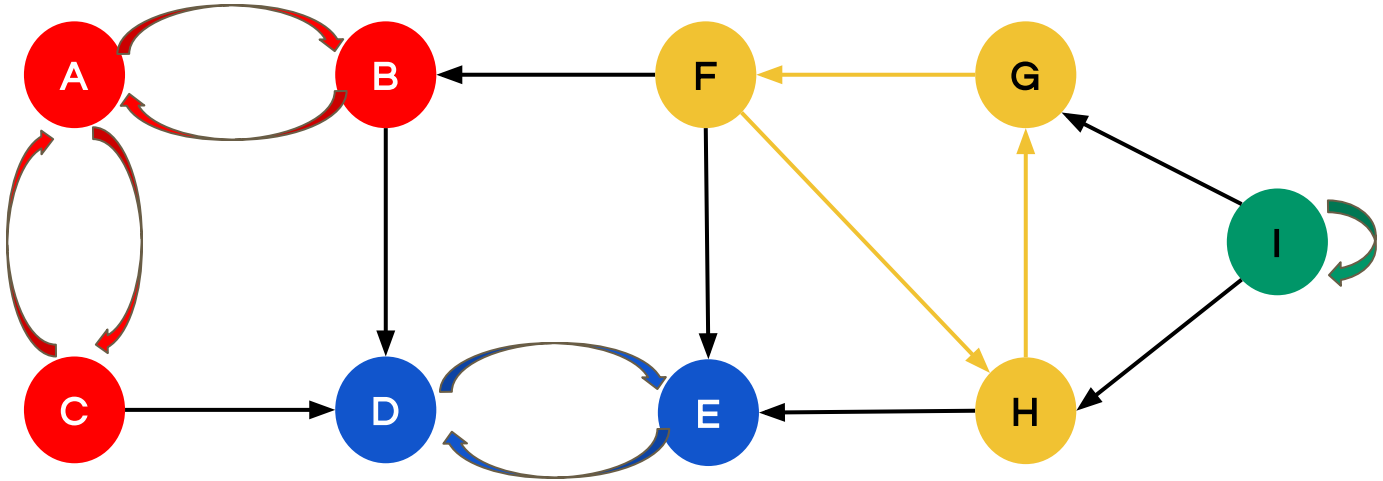


Applications of Depth First Search

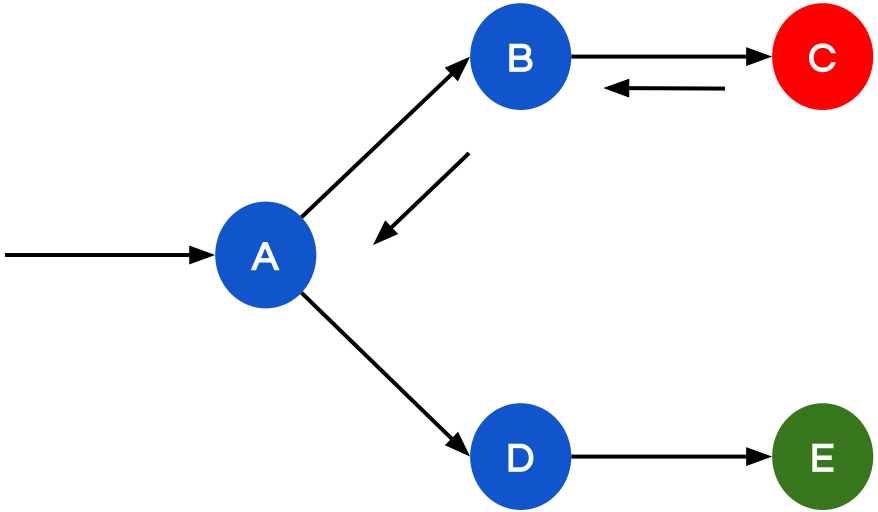
Detecting Cycle in a Directed Graph



Finding Strongly Connected Components of a graph



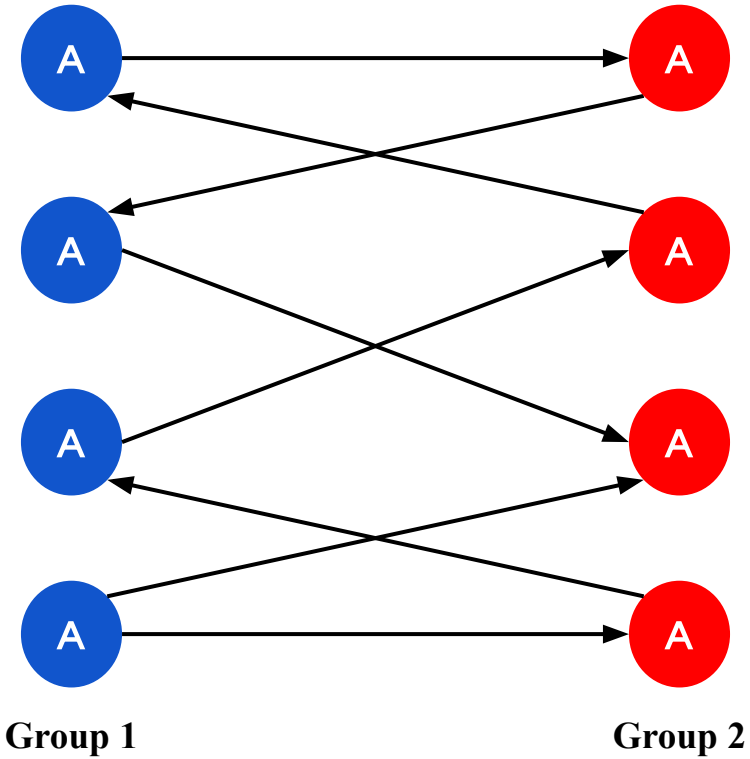
Backtracking



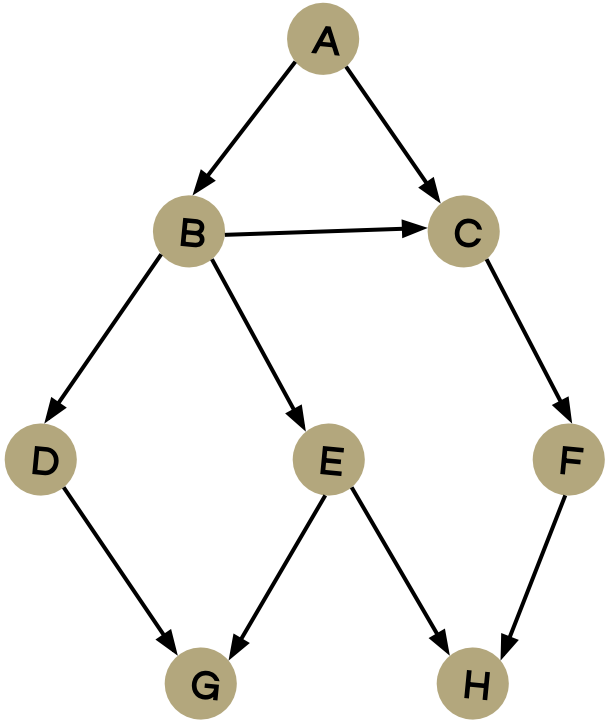
Not a solution

Solution

To test if a graph is Bipartite

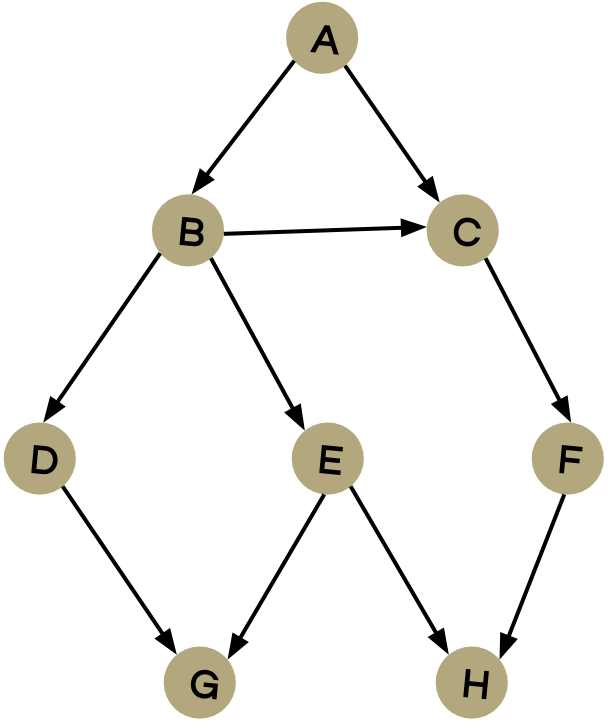


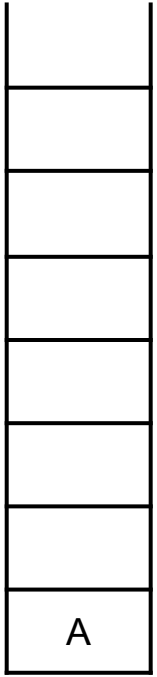
Depth First Search(1)



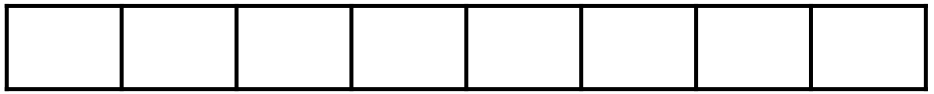
Vertex	Adjacent List
A	B, C
B	C, D, E
C	F
D	G
E	G, H
F	H
G	
H	

Depth First Search(2)



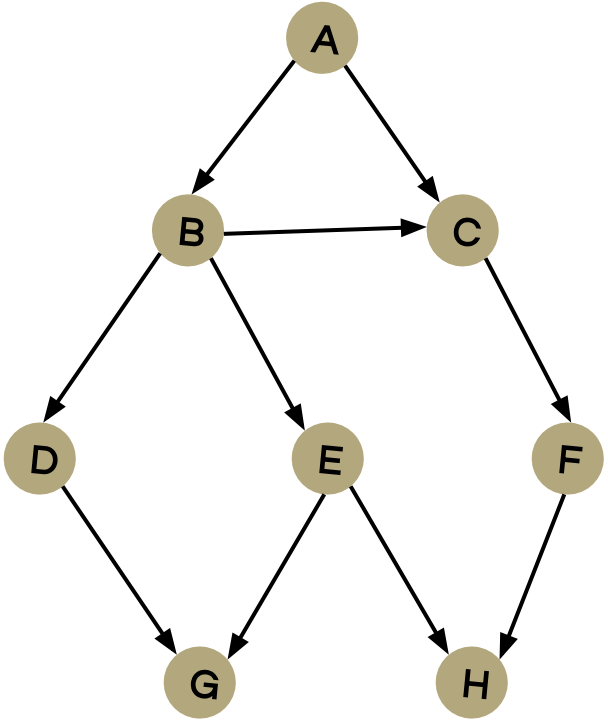


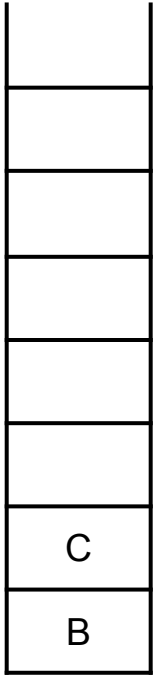
STACK



VISITED ARRAY

Depth First Search(3)



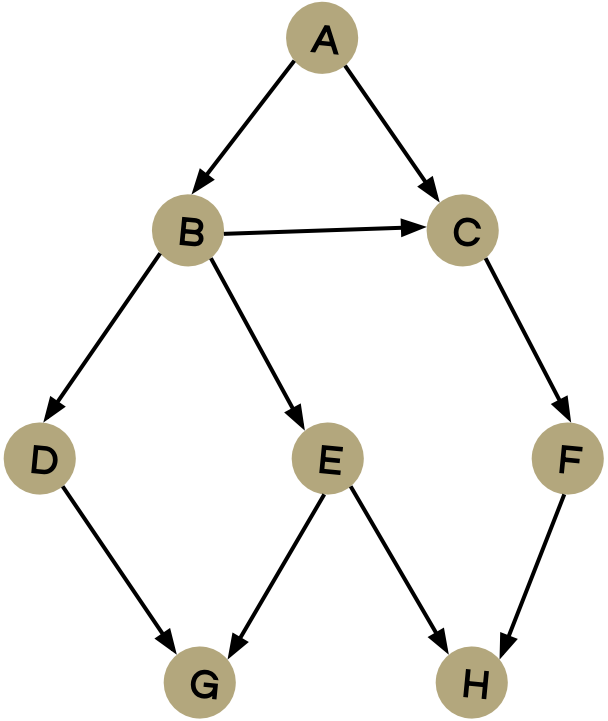


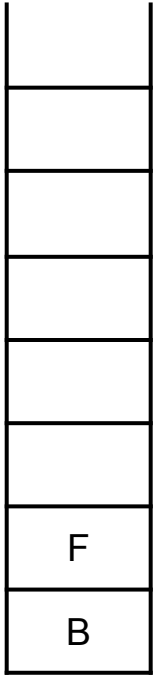
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VISITED ARRAY

Depth First Search(4)



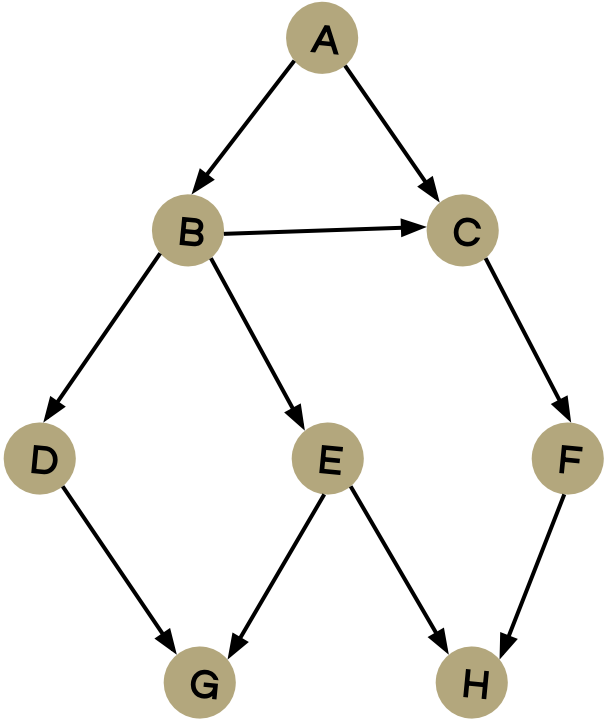


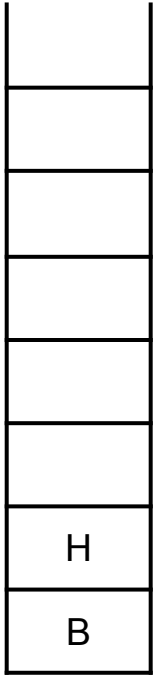
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A	C						

VISITED ARRAY

Depth First Search(5)



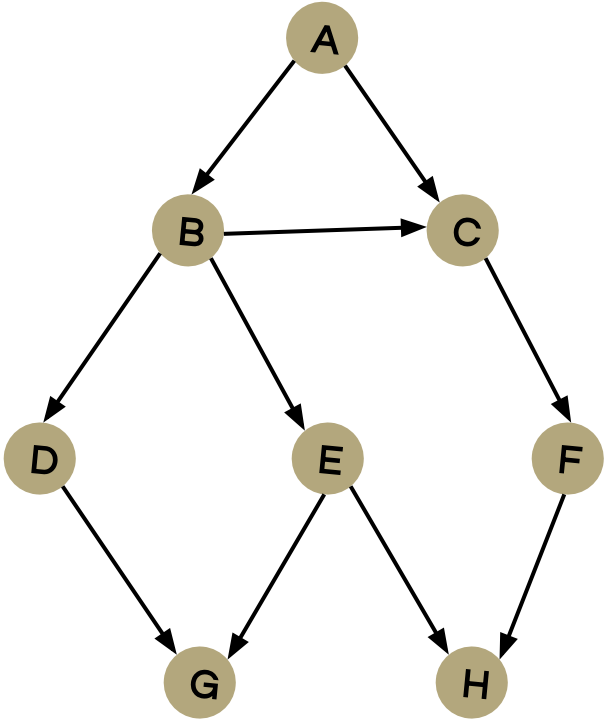


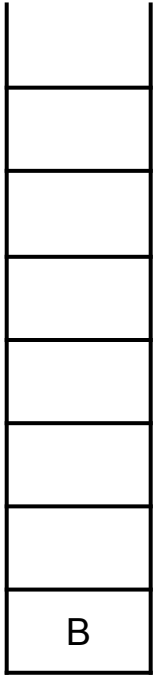
STACK

A	C	F					

VISITED ARRAY

Depth First Search(6)



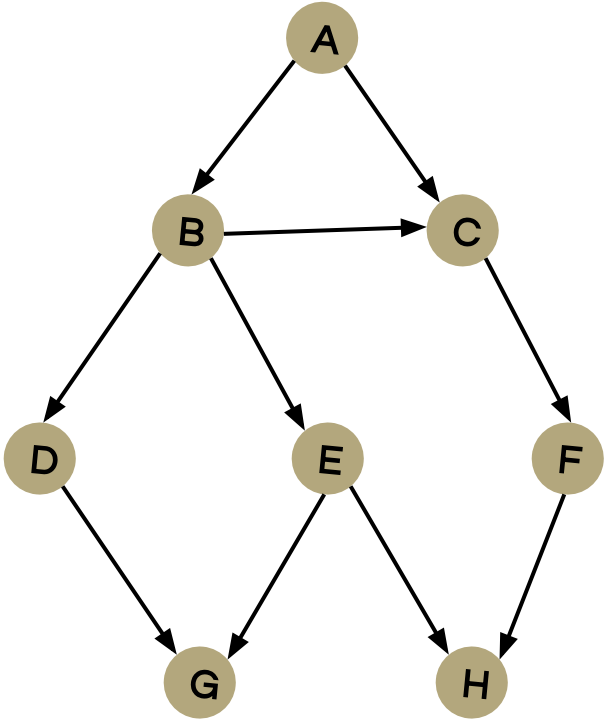


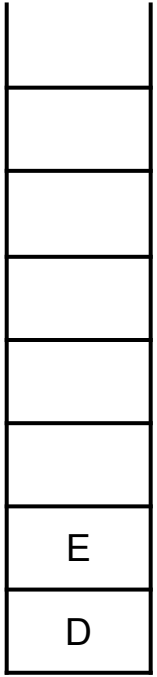
STACK

A	C	F	H				

VISITED ARRAY

Depth First Search(7)



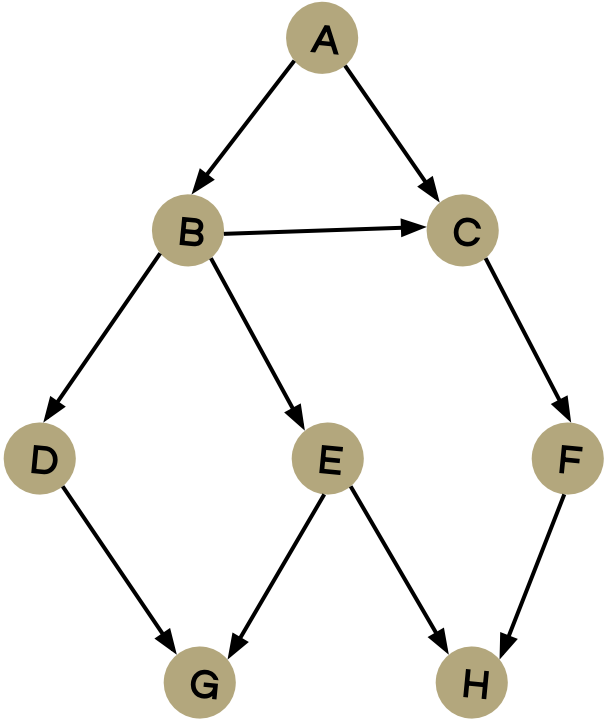


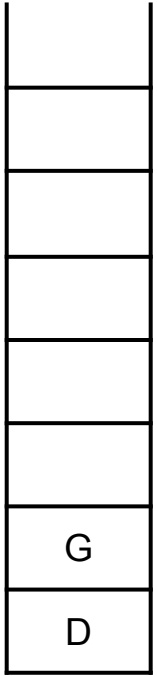
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A	C	F	H	B			

VISITED ARRAY

Depth First Search(8)



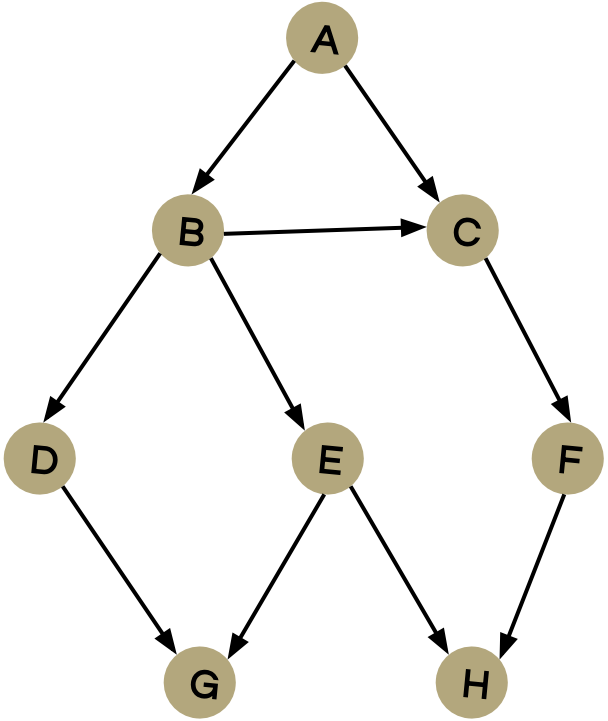


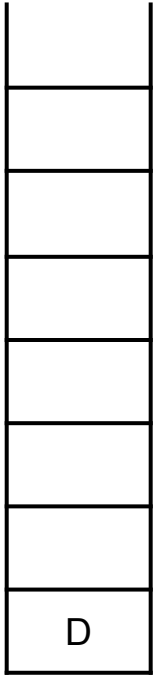
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A	C	F	H	B	E		

VISITED ARRAY

Depth First Search(9)



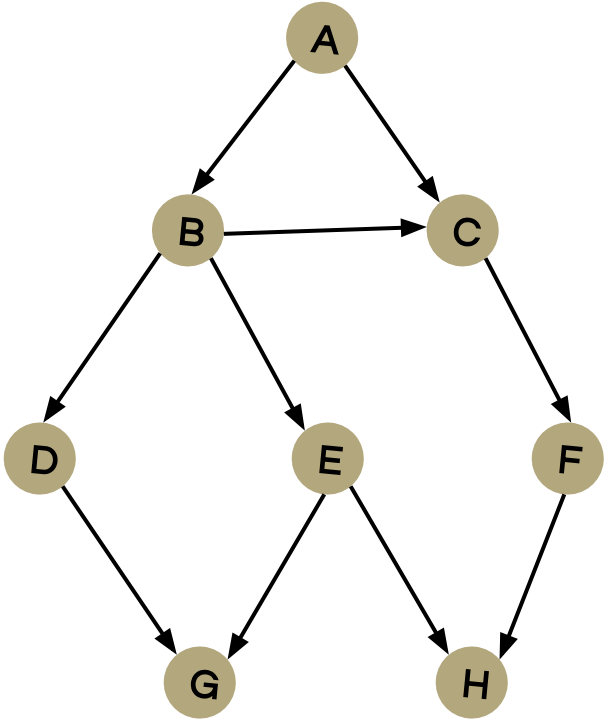


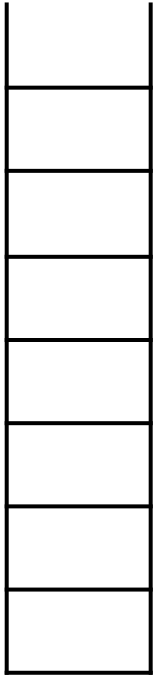
STACK

A	C	F	H	B	E	G	

VISITED ARRAY

Depth First Search(10)





STACK

A	C	F	H	B	E	G	D

VISITED ARRAY

Depth First Search

Disadvantages of DFS:

- Can get stuck exploring deep paths endlessly.
- No guarantee of finding a solution.
- No guarantee of optimal (shortest) solution.